	TOTAL FOR QUESTION 1	**8**
2(a)	The rate of change of momentum of a body is (directly) proportional to the resultant force acting on the body and is in the same direction as the resultant force.	**B1**
2(b)(i)	The sand initially has zero **horizontal** momentum but accelerates to the speed of the belt. Therefore, the belt exerts a forward horizontal force on the sand to produce this change in momentum. By Newton's third law, the sand exerts an equal and opposite horizontal force on the belt.	**M1** **A1**
2(b)(ii)	$\langle F \rangle = \frac{m}{t}\Delta v = (0.100)(0.050)$ $= 0.0050$ N	**M1** **A0**
2(b)(iii)1	$P = Fv = 0.0050 \times 0.050 = 2.5 \times 10^{-4}$ W	**B1**
2(b)(iii)2	In one second, rate at which sand gains kinetic energy, K:	

Qns	Answer	Marks
	$K = \frac{\frac{1}{2}m\Delta(v^2)}{t} = \frac{1}{2}\left(\frac{m}{t}\right)\left(v_f^2 - v_i^2\right)$ $= \frac{1}{2}\left(\frac{100}{1000}\right)\left(\left(\frac{5.0}{100}\right)^2 - 0\right)$ $= 1.25 \times 10^{-4}\,\text{W}$	B1
2(b)(iv)	Perfectly inelastic collision. Energy transferred mechanically via work done by the motor is to the (horizontal)l kinetic store of the sand and to the internal store of the system (heating/friction).	B1 B1
	TOTAL FOR QUESTION 2	8
3(a)	Electric potential at a point is the work done per unit charge by an external force in bringing a	B1